

EIED

Evidence Integrity & Eligibility Determination

Version 1.0 Portfolio Overview

A Healthcare AI Benchmark Suite for Evidence Validation, Documentation Integrity, and Eligibility Reasoning

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Has the AI earned the right to reach this conclusion?

Final v1.0 Portfolio Release | July 2026

github.com/reesemerres-jpg

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Executive Summary

Evidence Integrity & Eligibility Determination (EIED) is an independent healthcare AI benchmark suite designed to test whether available evidence actually supports a requested conclusion or decision. It focuses on a failure mode that is easy to miss: an AI response may sound clinically plausible, polished, and decisive while relying on stale, incomplete, contradictory, wrong-patient, or non-authoritative information.

EIED does not reward models for agreeing with a dashboard, trusting a signature, or escalating every ambiguous case. It requires the model to identify the claim, reconcile the full source record, judge the strength of the evidence, calibrate confidence, and choose only the decision the evidence earns.

Labels are claims—not evidence.

Version 1.0 contains twelve complete, multi-document benchmarks spanning discharge closure, prior authorization, clinical trial screening, population health registries, post-acute and procedural medical necessity, quality-measure validation, medication safety, case closure, clinical coding, longitudinal evidence reconciliation, and clinical guideline applicability. Each benchmark includes 30 fictional source documents, strict grading criteria, and design notes that document the intended failure modes and tempting shortcuts.

The suite is deliberately balanced. Some cases appear complete but are unsupported; others appear held or incomplete but are resolved by later authoritative evidence. This design punishes both “approve everything” and “escalate everything.”

Suite at a Glance

12 original benchmarks	360 fictional source documents	48 batch archives
12 strict grading rubrics	12 design notes	1 shared reasoning framework

CORE QUESTION Has the AI earned the right to reach this conclusion?

REASONING IDENTITY Claim → Evidence → Confidence → Decision

SIGNATURE PRINCIPLE Labels are claims—not evidence.

PRIMARY VALUE Tests evidence-supported decisions, not factual recall alone.

Core Philosophy

Trust must be earned from the evidence—not inferred from labels, signatures, dashboards, or assumptions.

- A signed document may still be internally inconsistent or contradicted by later evidence.
- An attached file may be irrelevant, wrong-patient, wrong-date, wrong-body-region, or outside the required window.
- An order, referral, or outreach attempt is not the same as completed action.
- A green or ready status is an assertion that must be validated.
- A hold is also an assertion and may be stale or resolved.
- Confidence should reflect the complete evidence set, including what remains unknown.

A conclusion should only be reached when the available evidence has earned it.

Claim → Evidence → Confidence → Decision

CLAIM	What status, conclusion, eligibility determination, or action is being asserted?
EVIDENCE	What current, patient-specific, internally consistent, and authoritative information supports or contradicts the claim?
CONFIDENCE	How strongly does the complete evidence set support the conclusion, and what remains unresolved?
DECISION	What action does the evidence justify—release, hold, correct, escalate, defer, include, exclude, finalize, revise, remove, or close?

Evaluation Pillars

Evidence Integrity Does the evidence actually support the claim?	Eligibility Determination Is the case eligible for the requested status or action?
Documentation Consistency Do the records agree, and are conflicts reconciled?	Temporal Integrity Did later information invalidate or resolve an earlier conclusion?
Source Reliability Which source is current, patient-specific, and authoritative?	Missing Evidence Is required proof absent, pending, attempted, or unverified?
Operational Readiness Can the workflow proceed once evidence requirements are satisfied?	Escalation Judgment Is release, hold, correction, escalation, or deferral proportionate?

Design Methodology

1. Start with a decision claim

Each benchmark begins with a status that appears simple: clean, ready, eligible, complete, safe, compliant, resolved, or supported.

2. Separate label from evidence

Dashboards and summaries are treated as claims. Source documents determine whether those claims are true.

3. Build object-specific source hierarchy

Current policies, patient-specific records, final addenda, specialist conclusions, helper files, and stale notes have different authority for different claim objects.

4. Add temporal, longitudinal, and provenance pressure

Later evidence may reverse earlier conclusions; repetition may create false durability; files may be wrong-patient, wrong-date, or outside the required window.

5. Validate applicability before invoking a rule

Policies, protocols, measures, and guidelines apply only after population entry, version, inclusion, exclusions, and safety gates are proven.

6. Include counterbalance cases

Every benchmark includes false-ready and false-hold cases so blanket approval and blanket escalation both fail.

7. Require decisions and grade the reasoning

Outputs must state confidence, disposition, owner actions, unresolved gaps, and the evidence that earns the decision.

Benchmark Index

Twelve domains. One shared question: does the evidence earn the decision?

#	Benchmark	Domain	Core decision
001	Discharge Documentation Closure Eligibility Audit	Clinical documentation integrity and care-transition closure	Has the record earned clean/complete closure status?
002	Prior Authorization Evidence Review	Payer criteria, imaging, specialty medication, and DME authorization	Has the request earned submission, hold, or escalation?
003	Clinical Trial Eligibility Validation	Oncology research screening and protocol eligibility	Has the candidate earned the right to proceed?
004	Registry Enrollment Audit	Population health registry and colorectal screening outreach	Has the patient earned inclusion, exclusion, suppression, or outreach?
005	Medical Necessity Documentation Review - Post-Acute Services	Home health, DME, wound therapy, oxygen, infusion, and outpatient therapy	Has the service earned a medically necessary determination?
006	Medical Necessity Documentation Review - Utilization Management	Imaging, procedures, treatment history, and exception pathways	Has medical necessity been established for this service and patient?
007	Quality Measure Inclusion Validation	Hypertension measure denominator, exclusion, and numerator review	Has the patient earned inclusion, exclusion, compliance, or hold status?
008	Medication Safety Verification	High-risk medication release, dispensing, and alert verification	Has the medication plan earned release or withholding?
009	Case Closure Readiness Audit	Patient safety, complaint, referral, privacy, communication, equipment, and follow-up cases	Has the case earned true resolution and closure?
010	Clinical Coding Evidence Validation	Diagnosis and principal-diagnosis coding integrity	Has the diagnosis or principal-diagnosis assignment earned final coding?
011	Longitudinal Evidence Reconciliation	Whole-record reconciliation across hospital, clinic, specialist, rehabilitation, home, pharmacy, and portal records	Does the longitudinal record support the conclusion?
012	Clinical Guideline Applicability Audit	Patient-specific guideline population, eligibility, exclusion, version, and safety-gate validation	Does the evidence actually justify applying this rule?

Each benchmark contains 30 fictional source documents, four batch archives, a task/system prompt, strict grading guidelines, and design notes.

BENCHMARK 001

Discharge Documentation Closure Eligibility Audit

Clinical documentation integrity and care-transition closure

Has the record earned clean/complete closure status?

PURPOSE Audits discharge records that appear complete and determines whether medication, wound, infection, follow-up, education, diagnosis, and DME evidence actually supports closure.

KEY EVIDENCE PRESSURE Signed-but-conflicting records; late results; wrong laterality; missing follow-up proof; stale copied-forward text; resolved apparent holds.

DECISION CLASSES Remain clean/complete; hold; escalate; defer; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE False-clean discharge records	COUNTERBALANCE PRESSURE Later authoritative evidence resolves stale holds
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What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 002

Prior Authorization Evidence Review

Payer criteria, imaging, specialty medication, and DME authorization

Has the request earned submission, hold, or escalation?

PURPOSE Tests whether prior authorization packets satisfy current payer criteria rather than merely appearing administratively complete or being labeled criteria met.

KEY EVIDENCE PRESSURE Criteria windows; body-region mismatch; therapy history; medication safety; exceptions; stale scratchpads; later addenda.

DECISION CLASSES Submit; hold; correct; withdraw; escalate; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE

Queue-ready requests without sufficient criteria evidence

COUNTERBALANCE PRESSURE

Later source evidence supports previously held requests

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 003

Clinical Trial Eligibility Validation

Oncology research screening and protocol eligibility

Has the candidate earned the right to proceed?

PURPOSE Evaluates protocol eligibility using consent version and timing, lab windows, washout periods, performance status, medication restrictions, imaging, language support, and exceptions.

KEY EVIDENCE PRESSURE Retired consent; pre-consent procedures; out-of-window labs; exact washout timing; current ECOG; interpreter attestation; final imaging.

DECISION CLASSES Eligible; hold; exclude; escalate to PI; defer; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE Tracker-eligible candidates who fail current protocol evidence	COUNTERBALANCE PRESSURE Final reports and protocol-specific addenda resolve apparent holds
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What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 004

Registry Enrollment Audit

Population health registry and colorectal screening outreach

Has the patient earned inclusion, exclusion, suppression, or outreach?

PURPOSE Audits registry labels against outside records, patient identity, procedure quality, screening intervals, high-risk factors, and stale exclusions.

KEY EVIDENCE PRESSURE Outside record matching; spouse-result mismatch; incomplete colonoscopy; scheduled vs completed; total vs segmental colectomy; stale hospice flags.

DECISION CLASSES Outreach; suppress; exclude; update; hold; escalate; release

FALSE-READY / FALSE-CLEAN PRESSURE

Incorrect overdue, complete, or exclusion labels

COUNTERBALANCE PRESSURE

Matched external evidence resolves missing-result and stale-hold cases

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 005

Medical Necessity Documentation Review - Post-Acute Services

Home health, DME, wound therapy, oxygen, infusion, and outpatient therapy

Has the service earned a medically necessary determination?

PURPOSE Determines whether post-acute service requests are supported by patient-specific skilled-need, functional, safety, and treatment evidence rather than signed forms or supplier-ready packets.

KEY EVIDENCE PRESSURE Template language; service-specific criteria; skilled need; homebound status; wound and oxygen proof; copied forms; later supporting notes.

DECISION CLASSES Support; hold; correct; escalate; deny/withdraw; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE Signed or supplier-ready packets without service-specific proof	COUNTERBALANCE PRESSURE Later documentation fully supports requests still carrying stale holds
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What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 006

Medical Necessity Documentation Review - Utilization Management

Imaging, procedures, treatment history, and exception pathways

Has medical necessity been established for this service and patient?

PURPOSE Reviews diagnostic and procedural requests for current symptoms, conservative treatment, body-region matching, procedure history, contraindications, and urgent exceptions.

KEY EVIDENCE PRESSURE Wrong body region; stale therapy; prescribed vs completed treatment; symptom-duration rules; identity mismatch; urgent exceptions; later addenda.

DECISION CLASSES Submit/approve pathway; hold; correct; withdraw; escalate; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE

Complete-looking packets that fail patient-, region-, timing-, or treatment-specific criteria

COUNTERBALANCE PRESSURE

Current examinations and authoritative outside evidence resolve apparent gaps

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 007

Quality Measure Inclusion Validation

Hypertension measure denominator, exclusion, and numerator review

Has the patient earned inclusion, exclusion, compliance, or hold status?

PURPOSE Separates denominator membership, valid exclusions, and numerator compliance rather than inheriting a compressed dashboard status.

KEY EVIDENCE PRESSURE Qualifying encounters; BP source rules; exclusion evidence; repeat readings; outside values; dates; stale status; denominator vs numerator conflation.

DECISION CLASSES Include; exclude; compliant; noncompliant; hold; release

FALSE-READY / FALSE-CLEAN PRESSURE

Dashboard classifications that collapse independent measure determinations

COUNTERBALANCE PRESSURE

Later valid evidence supports inclusion or compliance despite stale exclusions and holds

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 008

Medication Safety Verification

High-risk medication release, dispensing, and alert verification

Has the medication plan earned release or withholding?

PURPOSE Validates medication plans against current orders, renal function, weight, monitoring, outside prescriptions, recent events, discontinuations, and resolved allergy or duplicate alerts.

KEY EVIDENCE PRESSURE Stale labs; claims vs active therapy; critical late values; interacting prescriptions; cancelled procedures; overdose history; delabeling; outside monitoring.

DECISION CLASSES Release; hold; cancel; request information; urgent escalation; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE

Routine-looking refills and plans that are unsafe on current evidence

COUNTERBALANCE PRESSURE

Formal discontinuations, allergy testing, and matched labs resolve false alerts

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 009

Case Closure Readiness Audit

Patient safety, complaint, referral, privacy, communication, equipment, and follow-up cases

Has the case earned true resolution and closure?

PURPOSE Determines whether communication, corrective action, referral outcomes, source identity, privacy review, equipment resolution, and owner accountability support case closure.

KEY EVIDENCE PRESSURE Attempted vs completed communication; planned vs verified action; wrong-patient evidence; final vendor reports; privacy obligations; owner/due-date gaps.

DECISION CLASSES Close; hold; transfer; escalate; defer; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE Ready-to-close labels without completed resolution evidence	COUNTERBALANCE PRESSURE Matched final reports and confirmed resolution support closure despite stale holds
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What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 010

Clinical Coding Evidence Validation

Diagnosis and principal-diagnosis coding integrity

Has the diagnosis or principal-diagnosis assignment earned final coding?

PURPOSE Validates proposed coding against final patient-specific physician documentation, specialist conclusions, baseline trends, pathology, operative reports, addenda, and identity-correct evidence.

KEY EVIDENCE PRESSURE Abnormal lab vs diagnosis; empiric treatment; draft vs final documentation; baseline comparison; wrong-patient attachments; expected postoperative support; unsigned queries.

DECISION CLASSES Finalize; revise; remove; query; hold; escalate; release from hold

FALSE-READY / FALSE-CLEAN PRESSURE

Queue-ready diagnoses unsupported by final documentation

COUNTERBALANCE PRESSURE

Signed addenda, pathology, and operative evidence support held diagnoses

What strong performance looks like

- Identifies the surface claim without inheriting it as truth.
- Finds the decisive current and authoritative evidence across the full source set.
- Explains contradictions, missing proof, and confidence level.
- Makes a proportionate decision and releases resolved apparent holds.
- Assigns owner actions without inventing unsupported facts.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 011

Longitudinal Evidence Reconciliation

Whole-record reconciliation across hospital, clinic, specialist, rehabilitation, home, pharmacy, and portal records

Does the longitudinal record support the conclusion?

PURPOSE Determines whether a conclusion remains supported after the entire multi-month record is reconciled across hospital care, clinic follow-up, specialists, rehabilitation, home health, pharmacy, outside records, and patient communication.

KEY EVIDENCE PRESSURE Copied-forward diagnoses; repeated but unsupported causality; service discharge vs recovery; contextually unreliable measures; recurrence after closure; stale holds resolved by current evidence.

DECISION CLASSES Supported; supported with revision; not supported; unresolved; release from hold; keep open/hold

FALSE-READY / FALSE-CLEAN PRESSURE Repeated or copied conclusions appear established without whole-record support	COUNTERBALANCE PRESSURE Current functional, specialist, pharmacy, testing, and home evidence resolves stale restrictions
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What strong performance looks like

- Separates repeated documentation from independent confirmation.
- Reconstructs the chronology across settings without assuming the newest note automatically wins.
- Distinguishes condition, severity, cause, recovery, closure, eligibility, and safety as separate claims.
- Revises unsupported conclusions while releasing stale holds supported by current evidence.
- Assigns downstream problem-list, summary, restriction, closure, and follow-up corrections.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

BENCHMARK 012

Clinical Guideline Applicability Audit

Patient-specific guideline population, eligibility, exclusion, version, and safety-gate validation

Does the evidence actually justify applying this rule?

PURPOSE Determines whether patient-specific evidence establishes entry into the active guideline population and satisfies inclusion pathways, exclusions, setting, chronicity, and safety prerequisites.

KEY EVIDENCE PRESSURE Unconfirmed qualifying diagnosis; independent eligibility pathways; wrong guideline version; acute vs chronic abnormalities; excluded populations; disease eligibility vs medication-safety gates.

DECISION CLASSES Applicable; applicable with conditions; not applicable; insufficient evidence; defer pending verification; correctly withheld

FALSE-READY / FALSE-CLEAN PRESSURE Correctly quoted guidelines applied before patient-specific population entry is proven	COUNTERBALANCE PRESSURE Less familiar independent pathways support recommendations the dashboard denied
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What strong performance looks like

- Identifies the active guideline and proves population entry before applying the rule.
- Separates diagnosis, eligibility, exclusion, safety, and implementation confidence.
- Recognizes alternate eligibility pathways and avoids single-threshold denial.
- Rejects inapplicable rules while identifying the correct alternate pathway.
- Distinguishes future potential eligibility from current safe implementation.

Package: 30 source documents | 4 batch archives | system/task prompt | strict grading rubric | design notes

Cross-Benchmark Failure Taxonomy

Label trust	Treating green, complete, ready, compliant, safe, or closed as proof.
Signature trust	Assuming signed documentation is correct, sufficient, current, or internally consistent.
Temporal / longitudinal failure	Ignoring later evidence or allowing copied conclusions, recurrence, and stale statuses to survive whole-record review.
Provenance failure	Using wrong-patient, wrong-date, wrong-body-region, spouse, or copied-forward evidence.
Criteria / applicability failure	Replacing exact policy, protocol, measure, coding, population-entry, or safety requirements with a plausible shortcut.
Attempt-completion collapse	Treating ordered, scheduled, prescribed, contacted, or initiated as completed.
Source hierarchy failure	Allowing dashboards, scratchpads, drafts, or claims data to override final authoritative evidence.
Over-escalation	Holding or escalating everything instead of recognizing supported release and closure.
Under-escalation	Accepting surface completeness despite material evidence gaps or patient-safety risk.
Unsupported confidence	Presenting an uncertain conclusion as definitive without explaining remaining gaps.

Skills Demonstrated

- Healthcare workflow and documentation analysis across inpatient, outpatient, payer, research, population health, pharmacy, quality, coding, longitudinal care, and guideline-applicability domains.
- Multi-document source reconciliation and evidence hierarchy design.
- Creation of realistic synthetic records with temporal, provenance, identity, and criteria dependencies.
- Structured AI evaluation using explicit reasoning chains and strict grading rubrics.
- Failure-mode analysis, including tempting shortcuts and counterbalance cases.
- Translation of nursing judgment into reusable benchmark methodology.
- Clear portfolio documentation for evaluators, product teams, and hiring managers.

Relationship to HORB and UCM

HORB	EIED	UCM
Operational Reasoning <i>Can the AI reason through realistic healthcare operations?</i>	Evidence Integrity <i>Has the AI earned the right to reach this conclusion?</i>	Uncertainty Management <i>Has the AI exceeded what it can justify?</i>

Together, the suites form a portfolio framework for evaluating healthcare AI across operational reasoning, evidence-supported decision-making, and epistemic restraint.

Scope and Limitations

- All people, organizations, records, and scenarios are fictional and synthetic.
- EIED is an independent portfolio and evaluation-design project, not clinical, regulatory, payer, coding, legal, or research guidance.
- The suite has not been presented as a psychometrically validated public leaderboard benchmark.
- Performance should be interpreted against the provided task instructions and grading rubrics.
- Future work may include formal model testing, inter-rater review, scoring calibration, and versioned benchmark releases.

About Meredith Reese, RN

Meredith Reese is a Registered Nurse with experience across PCU, ICU, Neuro Trauma ICU, and hospital float nursing. Her benchmark work translates practical healthcare judgment—documentation review, prioritization, workflow constraints, evidence reconciliation, and interdisciplinary communication—into structured AI evaluation tasks.

Her healthcare AI portfolio includes HORB, focused on operational reasoning; EIED, focused on evidence integrity and decision justification; and UCM, focused on uncertainty and confidence boundaries.

GitHub: github.com/reesemerres-jpg

Version Information

VERSION 1.0

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CONTENTS 12 benchmarks, 360 fictional source documents, 48 batch archives, 12 prompts, 12 strict grading rubrics, and 12 design notes

Determine whether the evidence earns the decision.